

Case Study: End users in recovery from substance use disorders as designers and developers of digital games with therapeutic potential

Paula Maia de Souza
paulamaiasouza@gmail.com
Federal University of São Carlos
(UFSCar)
São Carlos, São Paulo, Brazil

Vinícius Matheus Romualdo
Santos
Federal University of São Carlos
(UFSCar)
São Carlos, São Paulo, Brazil
viniciusromualdo@estudante.ufscar.br

Bianca Alessandra de Souza
Alves
State University of Minas Gerais
(UEMG)
Passos, Minas Gerais, Brazil
biancaallessandra@gmail.com

Fernando Roberto Proença
Federal University of São Carlos
(UFSCar)
São Carlos, São Paulo, Brazil
fernandoroberto@gmail.com

Vivian Genaro Motti
George Mason University (GMU)
Fairfax, Virginia, United States
vmotti@gmu.edu

Kamila Rios da Hora Rodrigues
University of São Paulo (USP)
São Carlos, São Paulo, Brazil
kamila.rios@icmc.usp.br

Vânia Paula de Almeida Neris
Federal University of São Carlos
(UFSCar)
São Carlos, São Paulo, Brazil
vania.neris@ufscar.br

ABSTRACT

We propose a design process and a web system to support the design and development of digital games with therapeutic potential by people in recovery from Substance Use Disorders (SUDs). The proposal aims to generate more autonomy for end users as a means of scaling game design for therapy. This case study aims to evaluate the design process and the web system. We also assessed the autonomy of end users (patients). We employ a case study, as a methodological approach, conducting six meetings with design activities and the development of digital games with patients in recovery from SUDs and their therapists. Research data were collected through field diaries, observation, questionnaires, and interviews. Results suggest that participants were more autonomous with the support of the web system. This paper also features a discussion of the authors' perception of the activities, design implications, and recommendations for future research in the same domain.

CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods**; *Participatory design*; Empirical studies in interaction design.

KEYWORDS

Game design, vulnerable population, alcohol and drugs, therapeutic games.

ACM Reference Format:

Paula Maia de Souza, Vinícius Matheus Romualdo Santos, Bianca Alessandra de Souza Alves, Fernando Roberto Proença, Vivian Genaro Motti, Kamila Rios da Hora Rodrigues, and Vânia Paula de Almeida Neris. 2024. Case Study: End users in recovery from substance use disorders as designers and developers of digital games with therapeutic potential. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3613905.3637114>

1 INTRODUCTION

Substance Use Disorders (SUDs) affect 35 million people worldwide and cause more than 500,000 deaths yearly [6, 11]. The World Health Organization (WHO) predicts that by 2030, mental illnesses will be the leading disease burden worldwide. Given this projection, technological advances create opportunities for close collaboration between computing and mental health researchers [12]. Computing has made efforts to develop systems to support the therapy of people with SUDs. Among these systems are therapeutic digital games, potentially engaging patients in the therapeutic process. The design and development of games in this domain requires an active multidisciplinary team, which includes computer and health professionals with experience in the pathology that will be treated. The active participation of end users, patients, and caregivers is also desired [9, 10]. However, little is known in the literature about people in recovery from SUDs as designers and developers of therapeutic digital games [8, 9].

Based on this context, an ongoing Ph.D. research seeks to investigate ways to support the design and development of digital games with therapeutic potential by people in recovery from SUDs so that they have more autonomy, feel empowered in the creation of digital games with therapeutic potential, and can express themselves through the design and stories created for the games, without depending on the direct intervention of computing professionals. For this, the Ph.D. research proposed the construction of a framework that supports design and development activities by this population, providing more autonomy for end users. More autonomy in this context is considered the ability of end users to apply activities with the least possible intervention from computing professionals.

The framework is being built with the following composition: i) a design process considering the SUDs context; ii) a set of rules to create Conversational Agents (CAs) that support and guide end users in SUDs rehabilitation in the design and development process of digital games with therapeutic potential; and iii) a Web platform where end users in SUDs recovery can apply the design process (see Fig. 1), guided by a CA.

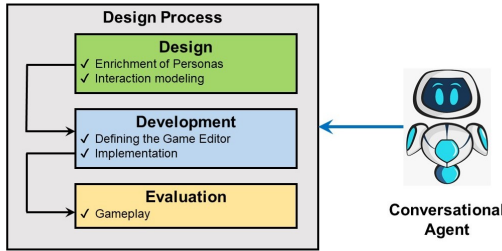


Figure 1: Suggested design process for SUDs context.

The Web platform proposed in the framework, henceforth SemTh Web [8], is in the evaluation phase. Therefore, this case study aimed to make SemTh Web available to professionals of a Psychosocial Care Center - Alcohol and Drugs (CAPS-AD, in Portuguese) to use with their patients. This case study aims to verify how much end users (patients) can use the system and evaluate the autonomy of these end users in designing digital games with therapeutic potential. The research hypothesis is that using the SemTh Web will give end users (patients in recovery from SUDs) more autonomy in designing digital games with therapeutic potential. This case study was approved by the Research Ethics Committee of the Federal University of São Carlos (in São Carlos city, State of São Paulo in Brazil) under feedback report (named CAAE) number 66623823.4.0000.5504.

2 METHOD

This research is characterized as an empirical case study, part qualitative and part quantitative. The case study has 6 stages, namely: plan, design, preparation, collection, analysis, and sharing [13].

2.1 Plan, design, and preparation of the study

The case study was chosen as a research method because it is a research that does not allow the control of behavioral events, seeks to answer questions about how participants would apply the proposed activities, and focuses on a contemporary event. The case

study also allows the collection of data from multiple methods, such as interviews, questionnaires, observation, and artifact analysis [13].

In the planning of the study, it was defined that it would be carried out in six meetings, which are detailed in Table 1. The number used as an identifier in the “ID” column of Table 1 refers to the number of meetings. In each meeting, participants carried out participatory activities to design digital games with therapeutic potential. For the activities, the steps proposed in the design process suggested for the SUDs context were followed (see Fig. 1).

Table 1: Activities planned for the study.

ID	Activity	Purpose	Data Collection Instruments
1	Presentation of the Project	Present the project to the participants; Signing of consent forms; Familiarize participants with the computer.	Field diary and observation.
2	Persona Enrichment Process	Identify, characterize, and profile potential players.	Field diary, observation, and content created.
3	Interaction modeling	Designing game scenes using predefined images.	Field diary, observation, and content created.
4	Game implementation - Part 1.	Creation of a first prototype of a digital game; Creation of scenarios, inclusion of game mechanics.	Field diary, observation, and content created.
5	Game implementation - Part 2.	Finalize the digital game; Creation of scenarios, inclusion of game mechanics.	Field diary, observation, and content created.
6	Evaluate of the games	Evaluate the gameplay.	Field diary, observation, and interview.

Research activities were carried out in spaces made available by CAPS-AD in Passos City, State of Minas Gerais in Brazil. The CAPS-AD is an open-door and daily care service aimed at the treatment and service users’ family, social, and community reintegration [5]. CAPS-AD provides daily care and therapeutic to people in recovery from SUDs. The service has the participation of multidisciplinary teams composed of social workers, psychiatrists, psychologists, occupational therapists, nurses, and nursing assistants, among others [7].

The target audience for this research consists of people in recovery from SUDs, participants in rehabilitation activities at CAPS-AD, and their respective therapists. A therapist is considered to be any CAPS-AD health professional who participates in the therapeutic process of service users.

The project was presented to CAPS-AD health professionals, and all who wished to participate were welcome. Health professionals who wanted to participate in the project were responsible for recruiting their patients. All CAPS-AD users interested in the study were welcome, and the inclusion criteria for research participants were:

- The participant was undergoing a therapeutic rehabilitation process for SUDs at the CAPS-AD in Passos City;
- The participant was invited and supervised by a CAPS-AD health professional;
- The participant was interested in participating in the study;
- The participant agreed to participate by signing the free and informed Consent Form.

2.2 Collection, analysis and sharing data

Field diaries, questionnaires, interviews, and artifacts generated during the design activities were used for data collection. The health and computing professionals who followed the activities kept field diaries of all six meetings. The participants (patients) initially answered a profile questionnaire and, at the end of the activities, answered the System Usability Scale (SUS) [2] questionnaire and an interview to evaluate the process.

The SUS is a standardized questionnaire for evaluating the usability of systems or products. The mean SUS score (68 points) was calculated to analyze the quantitative data obtained through the SUS questionnaire.

For the analysis of the qualitative data collected in the field diaries and interviews, the thematic analysis method [1, 3] was used. According to Corbin and Strauss [3], the thematic analysis method is a systematic technique for analyzing qualitative data that involves identifying and categorizing patterns in the data. The basic steps of thematic analysis are:

- (1) **Data preparation:** Before starting the thematic analysis, the qualitative data must be reviewed and organized. Data can be transcribed and edited to ensure it is easily readable and understandable;
- (2) **Identification of themes:** Identifying themes involves carefully and repeatedly reading qualitative data to identify ideas, concepts, or words that repeatedly crop up in various parts of the data. Topics should be identified openly and without bias to ensure that relevant information is not lost;
- (3) **Categorization of theme:** Identified themes are grouped into broad categories based on their similarities and relationships. Categories can be created from themes or can be identified during data analysis. Categories are often used as labels to help organize data;
- (4) **Pattern analysis:** Pattern analysis involves a careful review of categories to identify relationships and patterns that emerge between them. Researchers can use diagrams, maps, or other techniques to visualize relationships between categories;
- (5) **Validation of results:** Validation of results involves review and confirmation of analysis results by other researchers or study participants. This helps ensure the reliability and validity of the results;
- (6) **Report of results:** Analysis results are reported clearly and objectively, often with examples of data that support the conclusions. Reporting the results should include a detailed description of the themes and categories identified, as well as an analysis of the patterns that emerge among them.

Thematic analysis can be carried out manually or using some software for qualitative data analysis [1]. For the analysis carried out in this study, we used the ATLAS.ti¹. The evaluation results are described in Subsection 3.3.

With the end of the analysis of the data collected in the case study, we now have the opportunity to share the results obtained through this scientific paper. We therefore hope to contribute to the academic community and to future research in the same domain.

3 RESULTS

In this section, we present the details of the results of the case study. A computer researcher and two CAPS-AD health professionals participated in the six meetings held at the institution, which took place in April and May 2023. They invited their patients to participate in the activities. All who wanted to participate were instructed to sign the consent form and answer a profile questionnaire. A total of 25 patients participated in the activities. As CAPS-AD is an open-door institution, not all participants were always present at all meetings. However, the computer researcher already had experience in this scenario and anticipated this participant turnover. Some measures were adopted to mitigate potential losses that this turnover could cause in the research, such as dividing groups, presenting consent terms to new participants, and supporting the return to the train of thought explaining what has already been done and what should be done. The profile of the participants is described in the Subsection 3.1.

The participants were divided into two groups (G1 and G2). Each health professional accompanied a group. G1 used the SemTh approach with paper artifacts for game design. G2 used the SemTh Web system. One of the objectives was to assess how much more autonomous G2 was and how much less intervention was needed by the computer researcher who accompanied the activities. Details of the six meetings are described in the Subsection 3.2.

In all meetings, the computer researcher and the health professionals kept field diaries. In the last meeting, participants were instructed to answer the SUS questionnaire, and an interview was conducted about the activities carried out. The results of the analysis of the collected data are presented in Subsection 3.3.

3.1 Profile of participants

Of the 25 participants in the study, 22 participants (88%) identified themselves as men and three as women (12%). Age ranged from 18 to 60 years. In relation to the participants' education, the results of the profile questionnaire showed a variation from participants who did not complete basic education to participants with a complete technical level. Some participants attended CAPS-AD for less than a month (44%), and participants who attended for more than two years (24%).

Regarding the experience with computers, 12 participants (48%) reported having no experience and only 2 participants (8%) reported having a lot of experience. 23 participants (92%) said they had never participated in digital game creation activities and the same percentage also stated that they had never used any digital system to support their therapeutic process.

3.2 Activities carried out at CAPS-AD

In the first meeting, there were ten patients. The professionals presented the project to the patients and invited them to participate. All accepted and completed the consent term and the profile questionnaire. The first meeting aimed to present the project and provide participants with computers and digital games. The computer researcher took two laptops with two games developed in her research group. A game to teach the use of the mouse and another developed to encourage the self-esteem of people with depression.

¹<https://atlasti.com>

Some participants had the opportunity to use a computer for the first time at this meeting.

In the second meeting, the design activities began. Eight participants were divided into two groups (G1 and G2). In this meeting, both groups carried out the Persona creation activity. In this context, to facilitate understanding by the population, the activity was called defining the profiles of players. They defined the stakeholders and created the profiles and stories of potential players.

G1, accompanied by a health professional and a computer researcher, started the design using SemTh artifacts on paper. G2 went to another room accompanied by a health professional and created the design activities using the SemTh Web. Figure 2a shows a photo of G1 participants performing the activities and one artifact completed by G1. Figure 2b shows a photo of G2 participants performing the activities and one artifact completed by G2.

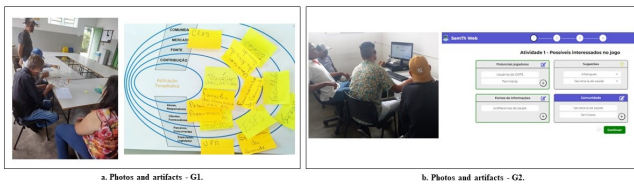


Figure 2: Second meeting - Photos and artifacts - G1 and G2.

In the third meeting, there were 14 participants. Participants who were not at the previous meeting were divided between groups. G1 continued doing the activities on paper and G2 using the SemTh Web. In this meeting, they were encouraged to carry out the interaction modeling activity. Within the context, the activity was applied in the format of providing a set of images of the real world for participants to create the game scenes (see Fig. 3). Participants were reminded that game scenes and stories should always reflect a way to help the player profiles defined in the previous meeting.



Figure 3: Third meeting - Photos and artifacts - G1 and G2.

To implement the games, a game engine called Lepi was used. Lepi supports the implementation of digital games by end users, not computer specialists [4]. The participants started using Lepi to implement the games in the fourth and fifth meetings. They remained divided into groups and were instructed to implement the scenes created in the third meeting, complementing them with dialogues, decisions, and punctuation. In the fourth meeting, there were 15 participants in the fifth meeting there were 13 participants.

G1 implemented 7 scenes in total. Figure 4a shows a sequence of scenes from the game created by G1. In this story, the character decides to leave home, meets a friend, and is thinking about going

to play ball. The player can then decide whether to play ball or drink alcohol. If he chooses to play ball that choice would lead to a positive consequence, when he chooses to go drinking the choice leads to a scene with a negative consequence.

G2 implemented 10 scenes in total. Figure 4b shows a sequence of scenes from the game created by G2. In this story, the main character's sister and girlfriend are angry with him for spending the night on the street and ruining his and their lives. The character says that he is fine and that they care a lot. The player can then make a choice between saying they are better off alone, continuing their life the same way, or accepting help. Each choice leads to other scenes with possible consequences.

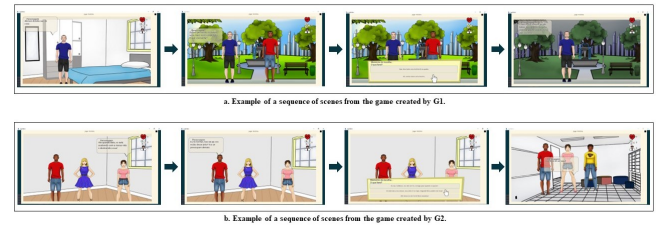


Figure 4: Examples of a sequence of scenes from the games created by G1 and G2.

In the sixth meeting, with the groups together, the participants were invited to play the game their colleagues had created. After the round of games, they were asked to respond to the SUS questionnaire and an interview to evaluate the activities. There were 5 participants in this meeting.

3.3 Data analysis

The result of the SUS questionnaire indicated a result of 69.5 points. This suggests that, according to the method, the evaluated design process had good usability acceptance by the research participants. Among the SUS results, in general, participants indicated that they felt confident and found it easy to use the process. They also pointed out that they would need support to use the process, but that they would learn quickly. They also indicated that they would like to use the process frequently.

To analyze the field diaries, the thematic analysis method was used. Firstly, the data preparation stage was carried out, in which the field diaries were transcribed into the ATLAS.ti. By reading the diaries, we identified themes and included them in the coding of the ATLAS.ti. The themes identified were grouped into 18 categories, being them: 1) Anxiety; 2) Autonomy; 3) Collaboration; 4) Creativity; 5) Difficulty; 6) Effectiveness in completing the task; 7) Empowered; 8) Engagement/interest of health professionals; 9) Engagement/interest of participants; 10) Intervention/help from health professionals; 11) Intervention/help from the computer researcher; 12) Satisfaction; 13) Resistance/Fear; 14) Resumption of the train of thought; 15) Rotation of participants; 16) Self-report; 17) Suggestions for process improvements; and 18) Therapeutic practices. The texts from the field diaries were analyzed and coded according to the theme categories.

Figure 5a shows the graph generated in ATLAS.ti after the qualitative analysis carried out in the field diaries in relation to G1. Figure

5b presents the graph generated in ATLAS.ti after the qualitative analysis carried out in the field diaries in relation to G2.



Figure 5: Graphs of qualitative analysis of field diaries from G1 and G2.

Observing the results of the data analysis in both groups, it was possible to identify that the participants and the health professional in G2, who used the SemTh Web, were more engaged in the activities. In the results, there were 31 occurrences of the theme “engagement/interest of participants” in G1, versus 50 occurrences of the same theme in G2. The theme “engagement/interest of health professionals” appeared in 15 occurrences in G1, versus 33 occurrences in G2.

Analyzing the results, it is also possible to observe that there was a difference in autonomy between the two groups. There were found 12 occurrences of the theme “autonomy” in G1 data, versus 28 occurrences in G2 data. G1 needed the researcher’s intervention 26 times, while G2 needed the researcher only 15 times. In the field diary of the health professional who accompanied G2 in meeting 4, the first meeting using SemTh Web, the professional reported that “*the help of the computer researcher was not necessary*”.

Another relevant difference is the collaboration between participants. In G1 there were 9 occurrences of the theme “collaboration”, while in G2 there were 24 occurrences of the same theme. According to the field diary of the health professional who accompanied G2 “*the group climate was creative, promoted identification, projection and reflection, collaborative, welcoming and sometimes relaxed postures even in the case of complex scenes*”.

In the G2 field diaries there were also more occurrences of the themes “creativity”, “empowered”, “satisfaction”, “self-report” and “therapeutic practices”. In contrast, in G2 there were fewer occurrences of the theme “difficulty”.

Two other themes considered very relevant in studies with people recovering from SUDs are the need to remember and help resumption the train of thought in sequential processes and the difficulty encountered in relating to turnover. In G1 there were 2 occurrences of the theme “resumption of the train of thought” and in G2 there were 4 occurrences of the same theme. This result may be related to the rotation of participants in the activities, appearing in G1 6 occurrences of the theme “rotation of participants” versus 12 occurrences in G2.

Only one mention of CA appeared in the field diaries. The health professional who accompanied G2 reported that “*P11 quickly understood that for each action he had to click on the ‘little robot’ SemTh*”. The ‘little robot’ mentioned referred to the avatar of SemTh Web’s CA. As only this mention appeared, it was not possible to evaluate how much the CA support helped G2.

In line with the other data analyzed, the interviews indicated that the participants enjoyed carrying out the activities. P22 reported that “*the experience was cool, I would like to repeat the meetings*”. In general, they reported that what they liked most was the interaction with their group colleagues and the fact that they could use the computer.

All participants who responded to the interview also agreed that the activities carried out helped in the therapeutic process. P16 reported that the activities “*encouraged me to use the computer, helped my head, got rid of bad thoughts and calmed me down a lot*”. P20 reported that “*it helped a lot because you interact with people, make new friends*”.

4 DISCUSSION

The design process presented to research participants brings together activities with characteristics adapted to the context. The process includes digital game design activities with therapeutic potential in the storytelling format, in which participants can express their opinions in the construction of stories for the games, reflecting on decisions related to real life. This game construction format makes participants reflect on their own daily choices, helping with rehabilitation, therapy and harm reduction related to SUDs. This section features a discussion of the authors’ perception of the activities, design implications, and recommendations for future research in the same domain.

Engagement and autonomy. Before we started the activities, health professionals mentioned that CAPS-AD users did not engage in activities for more than 40 minutes. However, all project meetings lasted between 1 hour and 15 minutes to 1 hour and 30 minutes. Furthermore, G2 participants only stopped activities because they were at lunchtime or were interrupted by health professionals because they had some other important activity. This result already indicates the adherence and engagement of participants in the activities.

Our hypothesis that the use of SemTh Web would give end users (patients in recovery from SUDs) more autonomy in designing digital games with therapeutic potential was validated by the results obtained. Data analysis suggests that participants who used the SemTh Web system were more engaged in activities, had more autonomy, and needed less help from the computer researcher. The health professional who accompanied G2, who used SemTh Web, also seemed more engaged in the activities. He said in his final field diary that he “*would like to thank us for opening and delivering the meetings and reiterate the invitation for future partnerships with the academy*”.

Collaboration. The health professional who accompanied G1 reported that she observed collaboration between the participants and that they helped and encouraged each other to participate. In G2’s field diaries, it was observed that everyone created the stories together, respecting each person’s profile and knowledge, and what the group decided was implemented. The health professional who accompanied G2 highlighted the collaborative and welcoming attitudes. And the participants themselves reported in the interviews aspects of interaction with colleagues and the possibility of new friendship ties.

In both working groups, collaboration between participants was observed. However, it was observed in the data analysis that in G2

there was more collaboration than in G1. Therefore, the results suggest that the use of the SemTh Web system supported collaboration between participants.

Therapeutic practices. The activities carried out and the resources used provided inspiration for health professionals. They pointed out that the creation of stories can be used in therapeutic groups. The activities carried out provoked reflection on the use of technology in the therapeutic process, to elaborate on existential, family, and relationship issues with drugs. According to health professionals, was very rich the possibility for users to express their characteristic vocabulary, their slang, compare the virtual situation with the real one, and project the consequences of their choices. Activities and encouragement to use the computer were also considered relevant for social reintegration, from the use of social networks to reintegration into the job market.

Design implications. The results of the study suggest that the design process adopted and the design decisions made in the development of the SemTh Web [8] were satisfactory in terms of generating engagement and promoting autonomy in participants. More accessibility features should be explored in future work. Health professionals were concerned about ensuring that the process was accessible to users with different levels of impairment (psychic organization, difficulties in cognition, drowsiness effects of the medication, lack of glasses). Some of these issues have already been considered at SemTh Web, but others have been addressed externally, with the support of health professionals and computer researcher.

Recommendations for future research. We believe that the main recommendation for future research in the field of SUDs is the need to approach the population in a real-life scenario. All the learning and results of this study were only possible due to the researchers' engagement in meeting this population in a real SUDs rehabilitation scenario.

When working with this population it is also necessary to take into account the rotation of participants. Research must have the flexibility to be adapted to the rotation without losing scientific rigor. A limitation of this study was the final evaluation by the participants. The rotation of participants limited the evaluation. Only 1 participant from G2 was present at all meetings that used SemTh Web and only he responded to the SemTh Web ease of use perception questionnaire, limiting this part of the evaluation. Of the 25 participants who attended the meetings, only 5 were at the last evaluation meeting. For these reasons, for future research, the authors suggest that evaluations be conducted per meeting, minimizing the effects of the rotation of participants.

ACKNOWLEDGMENTS

We thank all research participants, health professionals, and CAPS-AD users. We also thank the institutions that partially financed this study, CAPES and FAPESP.

This study was financed in part by the *Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001*.

Grant #2022/15468-7, São Paulo Research Foundation (FAPESP).

REFERENCES

- [1] H. Russell Bernard, Amber Y. Wutich, and Dr Gery W. Ryan. 2016. *Analyzing Qualitative Data: Systematic Approaches* (2nd ed. edição ed.). Sage Publications, Inc, Los Angeles.
- [2] John Brooke. 2013. SUS: a retrospective. *Journal of usability studies* 8, 2 (2013), 29–40.
- [3] Juliet M. Corbin and Anselm C. Strauss. 2014. *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory* (4th ed. edição ed.). Sage Publications, Inc, Los Angeles.
- [4] Franco Eusébio Garcia, Roberta Pereira Brandão, Gabriel Cheban do Prado Mendes, and Vânia Paula de Almeida Neris. 2019. Able to Create, Able to (Self-)Improve: How an Inclusive Game Framework Fostered Self-Improvement Through Creation and Play in Alcohol and Drugs Rehabilitation. In *Human-Computer Interaction – INTERACT 2019 (Lecture Notes in Computer Science)*, David Lamas, Fernando Loizides, Lennart Nacke, Helen Petrie, Marco Winckler, and Panayiotis Zaphiris (Eds.). Springer International Publishing, Cham, 337–358. https://doi.org/10.1007/978-3-030-29381-9_21
- [5] Clarissa de Barros Lacerda and Marta Fuentes-Rojas. 2016. Significados e sentidos atribuídos ao Centro de Atenção Psicossocial Álcool e outras Drogas (CAPS AD) por seus usuários: um estudo de caso. *Interface - Comunicação, Saúde, Educação* 21 (Oct. 2016), 363–372. <https://doi.org/10.1590/1807-57622016.0060> Publisher: UNESP.
- [6] Stefan Olafsson, Byron Wallace, and Timothy Bickmore. 2020. Towards a Computational Framework for Automating Substance Use Counseling with Virtual Agents. In *Proceedings of the 19th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS '20)*. International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 966–974.
- [7] Idma Martins Rocha. 2017. A relação do usuário do CAPS-AD com a atenção primária à saúde. 18p. Trabalho de conclusão de Residência apresentado à Fundação Estatal Saúde da Família e Fundação Osvaldo Cruz – Salvador/BA.
- [8] Paula Maia de Souza. 2024. *A Framework for the Design of Digital Games with Therapeutic Potential by People in Recovery from Substance Use Disorders*. Doctoral Thesis in Computer Science, Department of Computing, Federal University of São Carlos - UFSCar, São Carlos, SP, Brazil. 127 pages.
- [9] Paula Maia de Souza, Vânia Paula de Almeida Neris, Fernando Roberto Proença, and Franco Eusébio Garcia. 2022. End-users in recovery from substance use disorders as designers of Personas and digital games with therapeutic potential. *Journal on Interactive Systems* 13, 1 (Oct. 2022), 243–256. <https://doi.org/10.5753/jis.2022.2539> Number: 1.
- [10] Paula Maia de Souza, Kamila Rios H. Rodrigues, and Vânia Paula A. Neris. 2019. SemTh: an approach to the design of therapeutic digital games. In *Proceedings of the 18th Brazilian Symposium on Human Factors in Computing Systems (IHC '19)*. Vitória, ES, Brazil, 1–11. <https://doi.org/doi.org/10.1145/3357155.3358440>
- [11] Terra. 2021. OMS estima que mais de 500 mil mortes anuais ocorrem por conta da dependência química, transtorno cujo tratamento é muitas vezes a internação. Disponível em: <<https://www.terra.com.br/noticias/oms-estima-que-mais-de-500-mil-mortes-anuais-ocorrem-por-conta-da-dependencia-quimica-transtorno-cujo-tratamento-e-muitas-vezes-a-internacao,63f247a800e75de110d73cad4590ee2bm1skk3sm.html>> Acessado em: 04 de novembro de 2022. <https://www.terra.com.br/noticias/oms-estima-que-mais-de-500-mil-mortes-anuais-ocorrem-por-conta-da-dependencia-quimica-transtorno-cujo-tratamento-e-muitas-vezes-a-internacao,63f247a800e75de110d73cad4590ee2bm1skk3sm.html>
- [12] Greg Wadley, Rafael Calvo, John Torous, and Mary Czerwinski. 2018. 3rd Symposium on Computing and Mental Health: Understanding, Engaging, and Delighting Users. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems (CHI EA '18)*. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3170427.3170665>
- [13] Robert K. Yin. 2015. *Estudo de Caso: Planejamento e Métodos*. Bookman Editora.